



Lattice Theory for Parallel Programming

Solutions for exercises

1 Theoretical Exercises

Exercise 1

Show that if $(X, \leq)$ is a poset, then $(X, \leq^\partial)$ is also a poset. **Solution 1**

Let $x, y, z \in X$

1. $x \leq^\partial x \Leftrightarrow x \leq x$
2. $x \leq^\partial y \wedge y \leq^\partial x \Leftrightarrow y \leq x \wedge x \leq y \implies x = y$
3. $x \leq^\partial y \wedge y \leq^\partial z \Leftrightarrow y \leq x \wedge z \leq y \implies z \leq x \Leftrightarrow x \leq^\partial z$

Therefore, if $(X, \leq)$ is a poset, then $(X, \leq^\partial)$ is also a poset.

Exercise 2

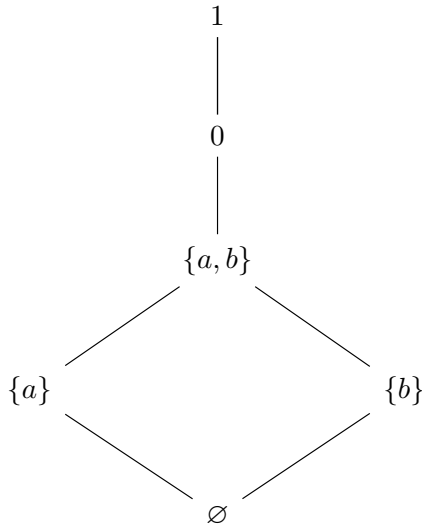
Show that $(\mathbb{P}(X), \Rightarrow)$ is a poset. **Solution 2**

Let $P, Q, R \in \mathbb{P}(X)$

1. $P \Rightarrow P \Leftrightarrow \{x \in X \mid P(x)\} \subseteq \{x \in X \mid P(x)\}$
2. $P \Rightarrow Q \wedge Q \Rightarrow P \Leftrightarrow \{x \in X \mid P(x)\} \subseteq \{x \in X \mid Q(x)\} \wedge \{x \in X \mid Q(x)\} \subseteq \{x \in X \mid P(x)\} \Leftrightarrow \{x \in X \mid P(x)\} = \{x \in X \mid Q(x)\} \Leftrightarrow P = Q$
3. $P \Leftarrow Q \wedge Q \Leftarrow R \Leftrightarrow \{x \in X \mid P(x)\} \supseteq \{x \in X \mid Q(x)\} \wedge \{x \in X \mid Q(x)\} \supseteq \{x \in X \mid R(x)\} \Leftarrow \{x \in X \mid P(x)\} \supseteq \{x \in X \mid R(x)\} \Leftrightarrow P \Leftarrow Q$

Exercise 3

If $(X, \leq_X)$ and $(Y, \leq_Y)$ are chains, then their linear sum $(X \oplus Y, \leq)$ is also a chain.

Solution 3**Exercise 4**

If $(X, \leq_X)$ and $(Y, \leq_Y)$ are chains, then their linear sum $(X \oplus Y, \leq)$ is also a chain.

Solution 4

Since $(X, \leq_X)$ and $(Y, \leq_Y)$ are chains, it means that all elements in each of which are comparable.

Also, by linear sum, all elements of X have to be placed below all elements of Y , while preserving the original orders within X and Y . Therefore, $\forall x \in X, \forall y \in Y, x \leq y$.

So, $(X \oplus Y, \leq)$ is a chain.

Exercise 5

The pointwise order of two posets is an order.

Solution 5

Let $(X, \leq_X)$ and $(Y, \leq_Y)$ be two posets. Let $(x_1, y_1), (x_2, y_2), (x_3, y_3) \in X \times Y$.

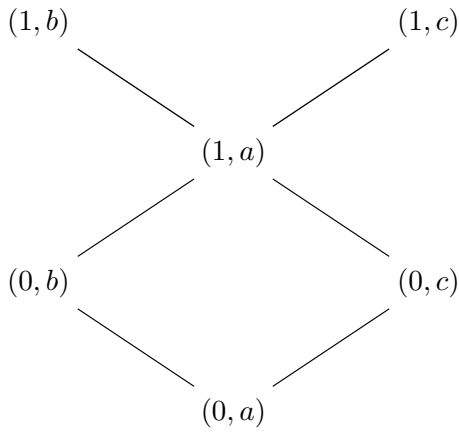
1. $(x_1, y_1) \leq (x_1, y_1) \Leftrightarrow x_1 \leq_X x_1 \wedge y_1 \leq_Y y_1$
2. $(x_1, y_1) \leq (x_2, y_2) \wedge (x_2, y_2) \leq (x_1, y_1) \Leftrightarrow (x_1 \leq_X x_2 \wedge y_1 \leq_Y y_2) \wedge (x_2 \leq_X x_1 \wedge y_2 \leq_Y y_1) \Leftrightarrow x_1 = x_2 \wedge y_1 = y_2 \Leftrightarrow (x_1, y_1) = (x_2, y_2)$
3. $(x_1, y_1) \leq (x_2, y_2) \wedge (x_2, y_2) \wedge (x_3, y_3) \Leftrightarrow (x_1 \leq_X x_2 \wedge y_1 \leq_Y y_2) \wedge (x_2 \leq_X x_3 \wedge y_2 \leq_Y y_3) \implies x_1 \leq_X x_3 \wedge y_1 \leq_Y y_3 \Leftrightarrow (x_1, y_1) \leq (x_3, y_3)$

Exercise 6

What is the Hasse diagram of $B \times A$ with lexicographic order?

Solution 6

$$B \times A = \{(0, a), (0, b), (0, c), (1, a), (1, b), (1, c)\}$$

**Exercise 7**

If $(X, \leq_X)$ and $(Y, \leq_Y)$ are chains, then the lexicographic order on product is also a chain. However, this is not necessarily true for the pointwise order on the product.

Solution 7

By definition of lexicographic order. If $(X, \leq_X)$ and $(Y, \leq_Y)$ are chains, then all elements in their product are comparable. Because it only depends on either $\leq_X$ or $\leq_Y$. However, in pointwise order, the elements in the product of $(X, \leq_X)$ and $(Y, \leq_Y)$ might be incomparable.

For example, $X = \{1, 2\}$, $Y = \{a, b, c\}$, their order is defined as natural and alphabetic.

Hence $X \times Y = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$, where $(1, b)$ and $(2, a)$ are incomparable.

Since $(1, b) \not\leq (2, a)$, $1 \leq_X 2$, but $b \not\leq_Y a$, $(X \times Y, \leq)$ is not a chain.

Exercise 8

Let $(X, \leq)$ and $(Y, \leq)$ be posets, and let $f: X \rightarrow Y$ be a function. The following conditions are equivalent.

1. f is an order isomorphism.
2. f is a monotone bijective map and the map f^{-1} is monotone.

Solution 8

- Assume (i) is correct. So f is an onto order embedding. It means that:

$$x_1 \leq x_2 \Leftrightarrow f(x_1) \leq f(x_2)$$

which is a monotone bijective map. $f^{-1}: Y \rightarrow X$, by (i), f is an onto function. Thus, every element in Y can be mapped by at least one element in X with $f(x)$. Also, f is one-to-one map, it means that 1 element in X will map to exactly one element in Y . By considering 2 properties, $\forall x_1, x_2 \in X$, $f^{-1}(f(x_1))$ and $f^{-1}(f(x_2))$ will map back to x_1 and x_2 , respectively.

- Assume (ii) is correct. Because f is a monotone bijective map, it is order-embedding. Also, when f^{-1} is monotone, it implies that every element in Y can find the corresponding element in X meaning f^{-1} is an onto. Therefore, by combining order-embedding and onto, we can conclude that f is order-isomorphism.

Exercise 9

Prove that the following statements are true.

1. The function $f: (\mathbb{N}, \leq) \rightarrow (\mathbb{N}, \leq)$ defined by $f(n) = 2n$ is order-preserving.
2. The function $g: (\mathbb{R}, \leq) \rightarrow (\mathbb{R}, \leq)$ defined by $g(x) = x^2$ is not order-preserving. For instance, $-2 < -1$, but $g(-2) = 4 > 1 = g(-1)$.
3. The inclusion map $i: (\mathbb{N}, \leq) \rightarrow (\mathbb{Z}, \leq)$ defined by $i(n) = n$ is an order-embedding.
4. The function $f: (\mathbb{N}, \leq) \rightarrow (\mathbb{N}, \leq)$ defined by $f(n) = 2n$ is an order embedding.
5. The function $f: (\mathbb{R}, \leq) \rightarrow (\mathbb{R}, \leq)$ defined by $f(x) = \lfloor x \rfloor$ is order-preserving but is not an order embedding.
6. The function $f: (\mathbb{N}, \leq) \rightarrow (\mathbb{N}^*, \leq)$ defined by $f(n) = n + 1$ is an order isomorphism.
7. The function $h: (\mathbb{N} \times \mathbb{N}, \leq) \rightarrow (\mathbb{N}, \leq)$, where $\mathbb{N} \times \mathbb{N}$ is equipped with the pointwise order, defined by $f(x, y) = x + y$ is order-preserving but not an embedding.
8. If $(X_1, \leq)$ and $(X_2, \leq)$ are two posets that the projections maps $\pi_1, \pi_2: X_1 \times X_2 \rightarrow X_i$ defined as $\pi_i(x_1, x_2) = x_i$ is order-preserving if $X_1 \times X_2$ is equipped with the pointwise order.

Solution 9

(1)-(4) and (6) are trivial.

1. Let $f(x) = 1, f(y) = 1$, then: $f(x) \leq f(y)$, but (x, y) can be $(1.5, 1.3)$ which means $x \not\leq y$. So, f is not order-embedding.
2. f is a monotone function. By pointwise-order, when $(x_1, y_1) \leq (x_2, y_2) \Leftrightarrow x_1 \leq x_2 \wedge y_1 \leq y_2$. So, $x_1 + y_1 \leq x_2 + y_2 \Leftrightarrow f((x_1, y_1)) \leq f((x_2, y_2))$. f is not order-embedding function. If $f((x_1, y_1)) \leq f((x_2, y_2)) \equiv 3 \leq 3$ (x_1, y_1) can be $(2, 1)$ and (x_2, y_2) can be $(1, 2)$. In such a case, $(x_1, y_1) \not\leq (x_2, y_2)$, so it is not order-embedding.
3. $\forall x_{11}, x_{12} \in X_1, \forall x_{21}, x_{22} \in X_2, (x_{11}, x_{12}) \leq (x_{12}, x_{22}) \Leftrightarrow x_{11} \leq x_{12} \wedge x_{21} \leq x_{22}$. So, by π_i , it will reserve one of dimensions, it means that the order relation will reserve the corresponding one.

Therefore, $\begin{cases} (x_{11}, x_{12}) \leq (x_{12}, x_{22}) \Rightarrow \pi_1((x_{11}, x_{12})) \leq \pi_1((x_{12}, x_{22})) & \Leftrightarrow x_{11} \leq x_{12} \\ (x_{11}, x_{12}) \leq (x_{12}, x_{22}) \Rightarrow \pi_2((x_{11}, x_{12})) \leq \pi_2((x_{12}, x_{22})) & \Leftrightarrow x_{21} \leq x_{22} \end{cases}$.

For lexicographic order, π_1 is order-preserving $\Leftrightarrow x_{21} \leq x_{22}$. Yet, π_2 is not.

Exercise 10

Let $X = \{a, b, c, d\}$. Prove that the powerset poset $(2^X, \subseteq)$ is isomorphic to the predicate poset $(\mathbb{P}, \Rightarrow)$. Then, prove that the previous statement holds for any set X (even an infinite one).

Solution 10

Define $\phi: 2^X \rightarrow \mathbb{P}(X)$. Let $g \in 2^X, \phi(g) = \{x \in X \mid x \in g\}$.

By this definition, ϕ is order-isomorphism, bijective. Therefore, $(2^X, \subseteq) = (\mathbb{P}(X), \Rightarrow)$

Exercise 11

Prove that $(\mathbb{Z}, \leq)$ is isomorphic to $(\mathbb{Z}, \leq^\partial)$. Is $(\mathbb{N}, \leq)$ isomorphic to $(\mathbb{N}, \leq^\partial)$?

Solution 11

We define $f: (\mathbb{Z}, \leq) \rightarrow (\mathbb{Z}, \leq^\partial), f(x) \triangleq -x$, such that $x_1 \leq x_2 \Leftrightarrow f(x_1) \leq^\partial f(x_2)$ Therefore, $(\mathbb{Z}, \leq) = (\mathbb{Z}, \leq^\partial)$.

Because, in $(\mathbb{N}, \leq)$, there is a bottom element, but top element. While, in $(\mathbb{N}, \leq^\partial)$, there is no bottom element but top element. So, we cannot find an order-isomorphism between them, they're not isomorphic.

Exercise 12

Characterize the $n \in \mathbb{N}$ whose divisor poset is isomorphic to $(2^{\{0,1\}}, \subseteq)$.

Solution 12

Prime number

Exercise 13

Prove that $(\mathbb{Z}, \leq)$ is isomorphic to $(\mathbb{N}, \leq^\partial) \oplus (\mathbb{N}, \leq)$.

Solution 13

Define $f: \mathbb{Z} \rightarrow \mathbb{N}$

$$f \triangleq \begin{cases} x - 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ |x|, & \text{if } x < 0 \end{cases}$$

Exercise 14

Let C be the set of subset X of $\mathbb{N}$ such that $\mathbb{N} \setminus X$ is finite. Show that C is a filter in $(2^{\mathbb{N}}, \subseteq)$.

Solution 14

Let $X \in C, Y \supseteq X$ such that $\mathbb{N} \setminus X$ is finite (by given condition). Then $\mathbb{N} \setminus X \supseteq \mathbb{N} \setminus Y$, therefore, $\mathbb{N} \setminus Y$ is also finite. It means that $Y \in C$. Hence, C is an up-set (upward-closed). In addition, for any $X, Y \in C$, there is $Z \in C$ such that $Z \subseteq X \wedge Z \subseteq Y$. This Z would be $\{\infty\}$. Assume that $X \wedge Y = Z, X \supseteq Z \wedge Y \supseteq Z$. $\mathbb{N} \setminus X$ is finite, $\mathbb{N} \setminus Y$ is finite, $\mathbb{N} \setminus Z = \mathbb{N} \setminus (X \cap Y) = \mathbb{N} \setminus X \cup \mathbb{N} \setminus Y$. Therefore, $C = \{X \in 2^{\mathbb{N}} \mid \{\infty\} \subseteq X\}$.

Exercise 15

Let $(X, \leq)$ be a poset and $Q \subseteq X$. We set

1. Show that $\downarrow Q$ is a down-set that contains Q . Deduce that $\uparrow Q$ is an up-set that contains Q .
2. Let $(Up(X), \supseteq)$ be the set of up-sets of $(X, \leq)$ ordered by reverse inclusion. Show that the map $f: X \rightarrow Up(X)$ defined as $f(x) = \uparrow x$ is an order-embedding.

Solution 15

$\downarrow Q = \{x \in X \mid \exists q \in Q, x \leq q\}$, by definition. Down-set means that if $y \in Y \wedge x \leq y$, then $x \in Y$, where $Y \subseteq X$. Since $Q \subseteq X \wedge Q \neq \emptyset$, meaning that $(Q, \leq)$ is a poset as well. By definition of $\downarrow Q$, since $(X, \leq) \& (Q, \leq)$ are posets, $\forall x \in X \exists q \in Q, x \leq q$, x can be q (reflexive) such that $\downarrow Q$ contains Q . Moreover, $\downarrow Q$ contains every element that is lower than at least one element in Q . It means that if we have an element $x \in X$ and an element $y \in \downarrow Q$ and their relation is $x \leq y$. Then it implies x must be in $\downarrow Q$, by the definition of $\downarrow Q$. Therefore, $\downarrow Q$ is a down-set. Dually, $\uparrow Q$ is an up-set that contains Q .

- Show that the map $f: X \rightarrow UP(X)$ defined as $f(x) = \uparrow x$ is an order-embedding.
- Order embedding: $x \leq y \Leftrightarrow f(x) \supseteq f(y)$.
- $\uparrow x = \{y \in X \mid y \geq x\}$.

($\Rightarrow$)

Since $x \leq y, |f(x)| \geq |f(y)|$. Also, by the definition of $\uparrow x$, $\uparrow x$ contains every element that above x (including y , and the elements above y). But x is below y , it means that $\uparrow y$ does not contain x . Still $\uparrow y$ contains every element that y . Hence $\uparrow y$ is a subset of $\uparrow x$ such that $\uparrow x \supseteq \uparrow y \Leftrightarrow f(x) \supseteq f(y)$.

($\Leftarrow$)

When $f(x) \supseteq f(y) \Leftrightarrow \uparrow x \supseteq \uparrow y$ is true. We can find the minimal elements x, y in $f(x), f(y)$, respectively. Both elements are in X such that $x \leq y$. Therefore, the map f is an order-embedding, it is an injective function.

Exercise 16

Prove that if $(X, \leq)$ has a top element, then it is unique. Similarly, if $(X, \leq)$ has a bottom element, then it is unique.

Solution 16

Suppose that we have 2 top elements, x, y , in $(X, \leq)$. By the definition of top element, $x \leq y$ and $y \leq x$ are the upper bounds of X . By the definition of upper bound, we know that $x \leq y$ and $y \leq x$. By antisymmetric, we can therefore conclude $x = y$. Therefore, if $(X, \leq)$ has a top element, it is unique. Dually, if $(X, \leq)$ has a bottom element, it is also unique.

Exercise 17

Prove that in any lattice $(L, \leq)$, for all $x, y, z \in L$:

1. $x \vee y = y \vee x$ and $x \wedge y = y \wedge x$ (commutativity)
2. $(x \vee y) \vee z = x \vee (y \vee z)$ and $(x \wedge y) \wedge z = x \wedge (y \wedge z)$ (associativity)
3. $x \vee x = x$ and $x \wedge x = x$ (idempotence)
4. $x \vee (x \wedge y) = x$ and $x \wedge (x \vee y) = x$ (absorption)

Solution 17

1. $x \vee y = \{x, y\}^u = y \vee x$
2. $(x \vee y) \vee z = \{x, y\}^u \vee z = \{\{x, y\}^u, z\}^u = \{x, y, z\}^u$, $x \vee (y \vee z) = x \vee \{y, z\}^u = \{x, \{y, z\}^u\}^u = \{x, y, z\}^u$
3. $x \vee x = \{x, x\}^u$
4. $x \wedge y \leq x \Leftrightarrow x \vee (x \wedge y) \leq x \vee x \Leftrightarrow x \vee (x \wedge y) \leq x$. By join definition, $x \leq x \vee (x \wedge y)$. Therefore, $x \vee (x \wedge y) = x$.

Exercise 18

Prove that if $(L, \vee, \wedge, 0, 1)$ is an algebraic bounded lattice, then in the corresponding order-theoretic lattice $(L, \leq)$, the element 0 is the bottom element and 1 is the top element.

Solution 18

By the definition of bounded lattice algebra $\mathcal{L} = (L, \vee, \wedge, 0, 1)$.

1. $\forall x \in L, x \vee 0 = x$, and $x \wedge 1 = x$.
2. $\forall x \in L, x \wedge 0 = 0$, and $x \vee 1 = 1$.

By connecting lemma, we know that when $x \vee 0 = x$, $x \wedge 0 = 0$, it implies that $0 \leq x$, since every element $x \in L$ is greater than 0. In other words, 0 is a lower bound for all $x \in L$. Therefore, 0 is the bottom element in L . Similarly, when $x \vee 1 = 1$, $x \wedge 1 = x$, it means that $x \leq 1$, since every $x \in L$ is lower than 1. It is saying that 1 is an upper bound for all $x \in L$. Therefore, 1 is the top element in L .

Exercise 19

Consider the following statements about lattice operations and constructions:

1. The disjoint union of lattices is a lattice.
2. The linear sum of lattices is a lattice.
3. The lexicographic order on the product of lattices might not be a lattice.
4. The pointwise order on the product of lattices is always a lattice, with join and meet operations computed pointwise.

For each statement, provide a proof or a counterexample to justify why the statement is true or false.

Solution 19

1. false, when $x \in P, y \in Q, x \& y$ are incomparable. Also, meaning $x \& y$ do not have the least ub and the greatest lb .
2. true, by definition of linear sum, every element can always find $lub \& glb$.
3. true, in lexicographic order, the second dimension might not comparable such that it might not be a lattice. So $lub \& glb$ are defined by pointwise join/meet. For example, $\mathbb{M}_3 \times \mathbb{Z}$, $lub \& glb$ do not exist!
4. true.

Exercise 20

If $(L, \wedge, \vee)$ is a lattice and S is a sublattice of L , then $(S, \vee, \wedge)$ is a lattice.

Solution 20

When S is a sublattice of L , by definition, $\forall x, y \in S, x \vee y \in S, x \wedge y \in S$. It implies that every pair of element in S has the least upper bound and the greatest lower bound which are defined in S . Therefore, S itself is a lattice.

Exercise 21

Let $(L, \vee, \wedge, 0, 1)$ be a bounded lattice and S be a sublattice of L .

1. Prove that if S is a 0-sublattice, then 0 is the bottom element of S .
2. Prove that if S is a 1-sublattice, then 1 is the top element of S .
3. Give an example of a sublattice that has a bottom element different from 0 and a top element different from 1.

Solution 21

(1) & (2) are similar to exercise 63.

(3)

Let L is a bounded chain, $\{0, 1, 2, 3\}$, where top is 3 and bottom is 0. There exists a sublattice $S = \{1, 2\}$. 1 is the bottom of S , but not 0–element in L , 2 is the top of S , but not 1–element in L .

Exercise 22

Let $f: L_1 \rightarrow L_2$ be a lattice isomorphism. Prove that f^{-1} is a lattice isomorphism.

Solution 22

$f^{-1}: L_2 \rightarrow L_1$. Let $x_1, y_1 \in L_1, x_2, y_2 \in L_2$. We know $f(x_1 \vee y_1) = f(x_1) \vee f(y_1), f(x_1 \wedge y_1) = f(x_1) \wedge f(y_1)$. Also, f is bijective and monotone. We have to show that

1. f^{-1} is lattice homomorphism
2. f^{-1} is bijective

[join]

$$f(x_1) \vee f(y_1) = f(x_1 \vee y_1) \Leftrightarrow f^{-1}(f(x_1)) \vee f^{-1}(f(y_1)) = f^{-1}(f(x_1 \vee y_1)) \Leftrightarrow f^{-1}(f(x_1)) \vee f^{-1}(f(y_1)) = f^{-1}(f(x_1 \vee y_1)).$$

[meet]

$$f(x_1) \wedge f(y_1) = f(x_1 \wedge y_1) \Leftrightarrow f^{-1}(f(x_1)) \wedge f^{-1}(f(y_1)) = f^{-1}(f(x_1 \wedge y_1)) \Leftrightarrow f^{-1}(f(x_1)) \wedge f^{-1}(f(y_1)) = f^{-1}(f(x_1 \wedge y_1)).$$

Since f is bijective, f^{-1} will be bijective as well.

Exercise 23

Prove that if $f: L_1 \rightarrow L_2$ is a bounded lattice isomorphism, then $f^{-1}: L_2 \rightarrow L_1$ is also a bounded lattice isomorphism.

Solution 23

Given that $f: L_1 \rightarrow L_2$ is a bounded lattice isomorphism. So, we know f is bounded lattice isomorphism and bijective. Hence, let $x_1, y_1 \in L_1, x_2, y_2 \in L_2$.

[join]

$$f(x_1 \vee y_1) = f(x_1) \vee f(y_1) \Leftrightarrow f^{-1}(f(x_1 \vee y_1)) = f^{-1}(f(x_1) \vee f(y_1)) \Leftrightarrow f^{-1}(f(x_1) \vee f(y_1)) = f^{-1}(f(x_1)) \vee f^{-1}(f(y_1)) \Leftrightarrow f^{-1}(f(x_1)) \vee f^{-1}(f(y_1)) \Leftrightarrow f^{-1}(x_2 \vee y_2) = f^{-1}(x_2) \vee f^{-1}(y_2)$$

[meet]

$$f(x_1 \wedge y_1) = f(x_1) \wedge f(y_1) \Leftrightarrow f^{-1}(f(x_1 \wedge y_1)) = f^{-1}(f(x_1) \wedge f(y_1)) \Leftrightarrow f^{-1}(f(x_1) \wedge f(y_1)) = f^{-1}(f(x_1)) \wedge f^{-1}(f(y_1)) \Leftrightarrow f^{-1}(f(x_1)) \wedge f^{-1}(f(y_1)) \Leftrightarrow f^{-1}(x_2 \wedge y_2) = f^{-1}(x_2) \wedge f^{-1}(y_2)$$

Since f is bijective, f^{-1} is bijective as well.

Exercise 24

Give an example of bounded lattices L_1 and L_2 and lattice homomorphism $f: L_1 \rightarrow L_2$ which is not a bounded lattice homomorphism.

Solution 24

Let $L_1 = \{\perp, a, b, \top\}$, where $a \parallel b$. Let $L_2 = \{0, 1, 2, 3\}$, where its order is defined as natural. f is defined as $f(\perp) = f(a) = 1, f(b) = f(\top) = 2$. In this case, two tops and bottoms from L_1 and L_2 are not mapped.

Exercise 25

Give a precise inductive definition of the interpretation of a (bounde) lattice term on a (bounded) lattice.

Solution 25

It is very similar with the operations used in L and the elements shown in L .

1. $\forall x \in X, t^L(x) = x$
2. $t^L(0) = 0, t^L(1) = 1$
3. $t(x, y) = x \vee y, t^L(x, y) = x \vee y, t(x, y) = x \wedge y, t^L(x, y) = x \wedge y$

Exercise 26

Any lattice satisfies the equation $(x \vee y) \vee z = x \vee (y \vee z)$. If $X = \{a, b\}$ then the lattice $(2^X, \cup, \cap)$ satisfies the equation $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$ while the lattice N_5 depicted Fig. 2 doesn't satisfy it.

Solution 26

$$a \vee (b \wedge c) = a \neq (a \vee b) \wedge (a \vee c) = c$$

Exercise 27

Let $\mathcal{K}$ be a class of (bounded) lattices and $t = s$ be a class of (bounded) lattice equations. Show that if every element of $\mathcal{K}$ satisfies $t = s$, then every element of $\mathbb{S}(\mathcal{K}), \mathbb{P}(\mathcal{K})$ and $\mathbb{H}(\mathcal{K})$ satisfies $t = s$.

Solution 27

$\forall L \in K: t^L(\vec{a} = s^L(\vec{a})), \text{ where } \vec{a} \in L.$

- sublattices: Let $subL$ as the set of some sublattices of L . $\forall SL \in subL: t^{SL}(\vec{b}) = t^L(\vec{b} = s^L(\vec{b}) = s^{SL}(\vec{b}))$. Therefore, $SL \in L$, meaning that to interpret $t^{SL}(\vec{b})$, we can use $t^L(\vec{b})$. Similar for s .
- Homomorphism image: Let $L_1, L_2 \in K$, $f: L_1 \rightarrow L_2$ is surjective homomorphism. $f(t^{L_1}(\vec{a})) = t^{L_2}(f(a_1), \dots, f(a_n))$. $t^{L_2}(f(\vec{a})) = f(s^{L_1}(\vec{a})) = s^{L_2}(\vec{a})$.
- Products: Let $(L_i)_{i \in I}$ be a family of lattices. $\pi_{i \in I} L_i$ is the set of all tuples $(a_i)_{i \in I}$ with $a_i \in L_i$, equipped with pointwise operators. $t^L(\vec{a}) = (t^{L_i}(\vec{a}_i))_i = (s^{L_i}(\vec{a}_i))_i = s^L(\vec{a})$

Exercise 28

Give an example of a non-distributive lattice whose every element has a complement, but there is at least one element which has two complements.

Solution 28

$\mathbb{M}_3$ or $\mathbb{N}_5$

Exercise 29

Show that the class of complemented bounded distributive lattices is not equational.

Solution 29

This is a complemental bounded distributive lattices. Yet, its sublattices are not, e.g., $\{\perp, a, \top\}$. In general, any chain in this lattice is not satisfied. Therefore, a does not have a complement.

Exercise 30

Prove that in a Boolean algebra, De Morgan's laws hold:

$$(x \vee y)' = x' \wedge y' \text{ and } (x \wedge y)' = x' \vee y'$$

Solution 30

$$\begin{aligned} (x \vee y) \vee (x' \wedge y') &= (x \vee y \vee x') \wedge (x \vee y \vee y') = (1 \vee y) \wedge (x \vee 1) = 1 \wedge 1 = 1 \\ (x \wedge y) \wedge (x' \vee y') &= (x \wedge y \wedge x') \vee (x \wedge y \wedge y') = (0 \wedge y) \vee (x \wedge 0) = 0 \vee 0 = 0 \end{aligned}$$

Exercise 31

Prove that in a Boolean algebra B , the complementation operation is a bounded lattice isomorphism between the underlying bounded lattice of B and its order dual. (*Hint*: Prove that (i) B^∂ equipped with the complementation operation from B is a Boolean algebra, (ii) the complementation operation $\iota: B \rightarrow B^\partial$ is a monotone map which is its own inverse.)

Solution 31**Exercise 32**

Let B be the set of subsets X of $\mathbb{N}$ that satisfy B is finite or $\mathbb{N} \setminus B$ is finite. Show that B is a Boolean algebra.

Solution 32

$$\begin{aligned}
b_1 \vee b'_1 &= \mathbb{N} = 1 \triangleq b_1 \cup b'_1 = b_1 \cup (\mathbb{N} \setminus b_1) = \mathbb{N} \\
b_1 \wedge b'_1 &= \emptyset = 0 \triangleq b_1 \cap b'_1 = b_1 \cap (\mathbb{N} \setminus b_1) = \emptyset \\
(b'_1)' &= (\mathbb{N} \setminus b_1)' = \mathbb{N} \setminus (\mathbb{N} \setminus b_1) = b_1
\end{aligned}$$

Exercise 33

We adopt the notation of the previous paragraphs.

1. Show that $\downarrow f$ is an ideal in $X \multimap Y$.
2. Show that $\downarrow f$ is a complete bounded lattice in its own right, with $\emptyset$ and f as bottom and top element, respectively.

Solution 33**Exercise 34**

Let $c: 2^{\mathbb{R}^\infty} \rightarrow 2^{\mathbb{R}^\infty}$ be defined as $c(X) = (\bigvee X]$.

1. Show that c is a closure operator.
2. Describe the set K of closed elements.
3. Describe the complete lattice operation on K .
4. Describe $c(x)$ in terms of closed elements for every $x \in 2^{\mathbb{R}^\infty}$.

Solution 34**Exercise 35**

Let $(L, \wedge, \vee, \perp, \top)$ be a bounded lattice, and X be an infinite set of variables. For any $n \geq 0$, let $T_n(X)$ the set of n -ary bounded lattice terms over X and set $T(X) := \bigcup_{n \geq 0} T_n(X)$. Define $c: 2^L \rightarrow 2^L$ by

$$c(A) = \{t^L(a_1, \dots, a_n) \mid n \geq 0, t \in T(X) \text{ and } a_1, \dots, a_n \in A\}.$$

1. Show that c is a closure operator.
2. Describe the set K of closed elements.
3. Describe the complete lattice operations on K .
4. Describe $c(A)$ in terms of closed elements for every $A \in 2^L$.

Solution 35**Exercise 36**

Show that the constructions of Table 1 define are mutually inverse.

Solution 36**Exercise 37**

Let $(X, \leq)$ be a poset. Define $U := \{\uparrow x \mid x \in X\}$ and $D := \{\downarrow x \mid x \in X\}$, as well as the maps $\alpha: D \rightarrow U$ and $\gamma: U \rightarrow D$ defined as $\gamma(\uparrow x) := \downarrow x$ and $\alpha(\downarrow x) := \uparrow x$. Prove that

$$(L, \leq) \xleftrightarrow[\alpha]{\gamma} (M, \sqsubseteq)$$

Solution 37**Exercise 38**

Let $(C, \leq)$ be a complete totally ordered set and $\rightarrow$ be the binary operation defined on C by $a \rightarrow b := \bigvee \{x \in C \mid a \wedge x \leq b\}$. Prove that for any $a \in C$, we have

Solution 38**Exercise 39**

(From Galois Connection to Closure Operator). Assume that

Solution 39**Exercise 40**

Identify the Galois connections among the Galois connections we have introduced so far.

Solution 40